Reactor Core Isolation Cooling

Chapter 2.7

OBJECTIVES

1. Identify the purpose of the Reactor Core Isolation Cooling (RCIC) system.

OBJECTIVES

2. Describe the purpose, function and operation of the following RCIC system major components:
    a. Steam Supply Isolation Valves
    b. Steam to Turbine Valve
    c. Turbine Trip and Throttle valve
    d. Turbine Governor valve
    e. Turbine
    f. Oil System
    g. Line Fill System
    h. Suppression Pool Suction Valve
    i. Condensate Storage Tank Suction Valve
    j. Pump
    k. Discharge Valve
    l. Full Flow Test Valve
    m. Flow Controller

OBJECTIVES

3. Describe the following flowpaths of the RCIC system:
   a. Steam Supply and Exhaust
   b. Injection Flow
   c. Test Flow
   d. Line Fill
   e. Barometric Condenser and Receiver
4. Identify the RCIC System parameters which affect the following:
   a. Automatic Initiation
   b. Automatic Isolation
   c. Automatic Turbine Trip
   d. Suction Path Transfer

OBJECTIVES

5. Describe how the RCIC system interrelates with the following systems/components:
   a. Condensate Storage Tank
   b. Main Steam System
   c. Condensate and Feedwater System
   d. Suppression Pool
   e. High Pressure Coolant Injection System
   f. Nuclear Steam Supply Shutoff System

Purpose

Provide makeup water to the reactor vessel for core cooling when:

- the main steam lines are isolated

or

- the condensate and feedwater system is not available.

RCIC Components

Steam Supply Isolation valves

- The two steam warm up isolation valves (MOV 47 and MOV 48)
  - normally open to maintain the steam lines warmed and ready for RCIC initiation.
- The Inboard steam isolation valve (MOV 41) is A.C. powered
  - maintained open in the standby line up.
- The Outboard steam isolation (MOV 42) valve is DC powered
  - closed in the standby line up.
- Both of the steam supply isolation valves open on a RCIC initiation signal.
- All four of the RCIC steam line valves close if there is a RCIC isolation signal

RCIC Components

Steam to Turbine Valve

- The steam supply shutoff valve (MOV 43) provides steam isolation for the RCIC turbine in the standby condition.
  - The valve receives an open signal on a RCIC automatic initiation (-38") and automatically closes on Level 8 (+56.5").

Turbine Trip and Throttle Valve

- The turbine trip and throttle valve (MOV 44) is upstream of the governor valve
  - provides for rapid turbine tripping under various conditions.
- Valve is open in the standby condition
  - can be used to throttle steam flow to the RCIC turbine if the governor valve fails open.

RCIC Components

Turbine Governor Valve

- The governor valve is controlled by an electro hydraulic system.
- The valve is opened by spring force and closed by RCIC turbine governor control oil pressure.
- Turbine speed is controlled by varying the operating oil pressure that is opposing the spring force.

RCIC Turbine

- The RCIC turbine is designed to accelerate from a cold standby condition to full load conditions within 30 seconds.
- The RCIC turbine is a non-condensing Terry turbine
  - Designed to operate with a steam supply pressure ranging from 150 psig to 1135 psig.

RCIC Components

RCIC Oil System

- A small gear pump is driven by the RCIC oil system worm gear.
- Oil is supplied to the turbine and pump bearings for lubrication
- Oil is supplied to the governor valve as control oil.
- In the normal operating speed range, a spring loaded pressure regulating valve recirculates excess oil to the pump suction
  - This limits oil pressure to approximately 15 psig.

RCIC Components

RCIC Barometric Condenser System

- Prevents leakage from the turbine shaft seals and turbine exhaust casing drain.
- May be manually started or it automatically starts on a RCIC system initiation.
- Steam leakage is collected in the barometric condenser from the:
  - turbine gland seals
  - turbine trip throttle valve stem
  - governor valve stem
- Non condensable gasses are exhausted to the suppression pool via the vacuum pump.
- The condensate pump cycles on high and low level signals when the system is in operation.
  - When the system is in standby, the barometric condenser is gravity drained to Clean Radwaste (CRW).

RCIC Components

RCIC Line Fill System

- The line fill pump takes suction from the RCIC CST suction line
- Discharges at 40 psig to the RCIC discharge line
- Maintaining the piping full
  - Minimizes the injection time for the RCIC pump
  - Prevents piping voids from forming that could cause water hammer.

RCIC pump

- A centrifugal pump designed to deliver 400 gpm to the reactor
  - Flow rate is equal to the boil off rate from the reactor 15 minutes after shutdown.

RCIC Components

RCIC Pump Discharge Paths

- The RCIC pump discharges through:
  - the system flow element
  - the discharge valves
  - the air operated check valve
- RCIC injects into the 'A' feedwater line
  - flow is distributed inside the reactor vessel by the feedwater spargers.
- A pump minimum flow line to the suppression pool
- A cooling line for turbine auxiliaries
- A full flow test line (shared by the HPCI System)

RCIC Components

RCIC Turbine Exhaust Path

- The RCIC turbine exhaust line routes steam from the RCIC turbine to the suppression pool below water level.
- Vacuum breakers are installed on the exhaust line
  - Prevent drawing suppression pool water into the line via a vacuum created by condensation of steam in the line following RCIC operation.
- The exhaust line is protected from over pressure by a RCIC turbine trip on high exhaust pressure.
- The high exhaust pressure trip is backed up by rupture diaphragms which relieve pressure to the RCIC room.
- The space between the exhaust rupture diaphragms has pressure switches that initiate a RCIC isolation.

RCIC Components

RCIC Pump Suction Path

- The system can take suction from the:
  - condensate storage tank (CST)
  - suppression pool
- The normal suction is from the CST on a line common with the HPCI System suction line.
- The suction swap from the CST to the suppression pool
  - Low CST level
  - High suppression pool level

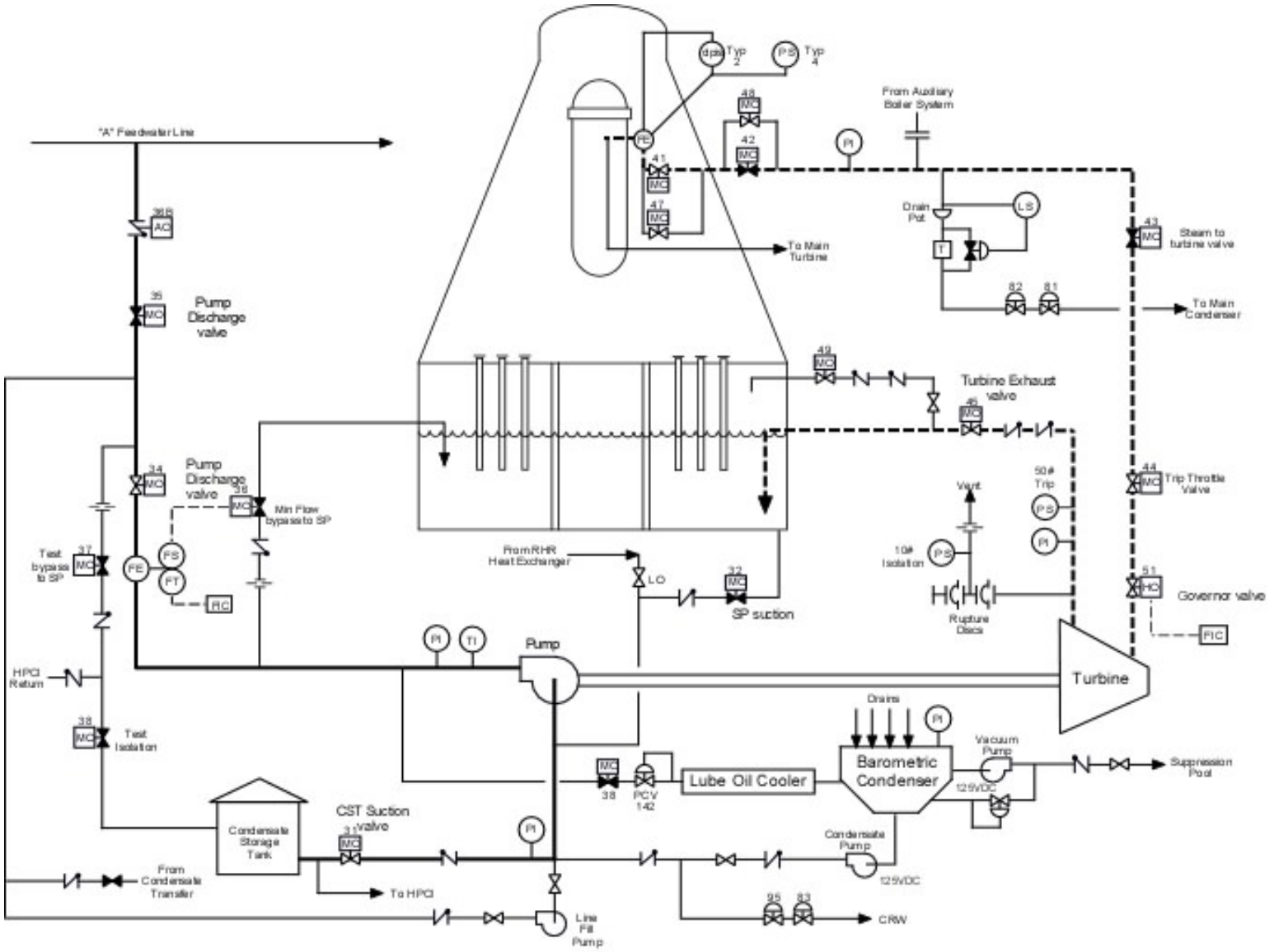


Figure 2.7-1 Reactor Core Isolation Cooling System

0601

RCIC Auto Operation

- RCIC initiates automatically at Level2 (-38”) or via the manual initiation pushbutton on P603.
  - RCIC steam to Turbine valve (MOV 45) closes at Level 8
  - RCIC will cycle automatically between L2 and L8 with no operator action

Turbine Trips

- Overspeed
  - Electrical (110%)
  - Mechanical (125%)
- Pump suction Pressure Low (15” Hg Vac)
- Turbine exhaust pressure high (50 psig)
- Any Isolation Signal
- Manual

Actions

- Closes Trip and Throttle valve

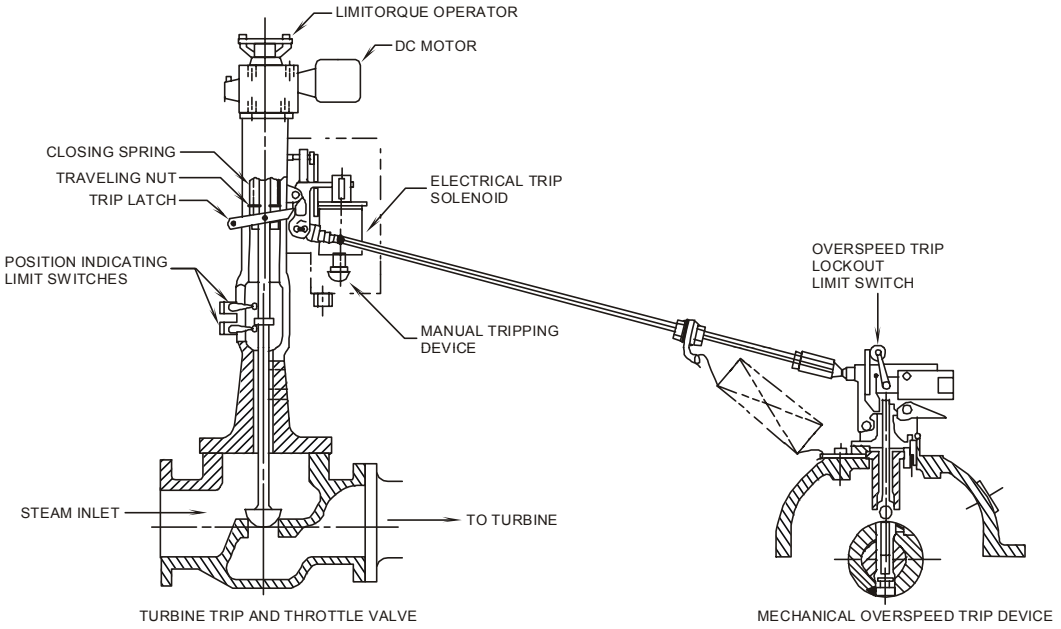


FIGURE 2.7-2 RCIC Turbine Trip and Throttle Valve

03 99

Isolations

- Steam Supply Pressure Low (57psig)
- High Steam Line Differential Pressure (291”H2O, 3 sec time delay)
- Steam Line Space High Temp (193 F)
- High Pressure between Rupture Discs (10 psig)
- Manual

Actions

- Inbd and Outbd Steam Isolations close (41, 42, 47, 48)
- RCIC turbine trips

System Interfaces

- Main Steam System (Section 2.5)
  - RCIC supplied from “A” steam line upstream of MSIVs
- Condensate and Feedwater (Section 2.6)
  - Water returns to vessel on FW line “A”
  - Steam drains are directed to Main Condenser
- High Pressure Coolant Injection (HPCI) (Section 10.1)
  - Back up to RCIC
  - Shared suction and test line from the CST

System Interfaces

- **Nuclear Steam Supply Shutoff System (Section 4.4)**
  - The NSSSS provides signals to isolate the RCIC Steam supply isolation valves
- CST
  - The primary source of water and test volume
- Suppression Pool
  - A secondary source of water and condenses steam

OBJECTIVE REVIEW

1. Identify the purpose of the Reactor Core Isolation Cooling (RCIC) system.

OBJECTIVE REVIEW

2. **Describe the purpose, function and operation of the following RCIC system major components:**
    a. Steam Supply Isolation Valves
    b. Steam to Turbine Valve
    c. Turbine Trip and Throttle valve
    d. Turbine Governor valve
    e. Turbine
    f. Turbine Exhaust valve
    g. Oil System
    h. Pump
    i. Minimum Flow Valve
    j. Flow Controller
    k. Suppression Pool Suction Valve
    l. Condensate Storage Tank Suction Valve
    m. Injection Valve
    n. Full Flow Test Valve
    o. Line Fill Pump
    p. Barometric Condenser

OBJECTIVE REVIEW

3. **Describe the following flowpaths of the HPCI system:**
   a. Steam Supply and Exhaust
   b. Injection Flow
   c. Test Flow
   d. Line Fill
   e. Barometric Condenser and Receiver
4. **Identify the RCIC System parameters which affect the following:**
   a. Automatic Initiation
   b. Automatic Isolation
   c. Automatic Turbine Trip
   d. Suction Path Transfer

OBJECTIVE REVIEW

5. **Describe how the RCIC system interrelates with the following systems/components:**
   a. Main Steam System
   b. Condensate and Feedwater System
   c. Suppression Pool
   d. Condensate Storage Tank
   e. Nuclear Steam Supply Shutoff System